

Generation Interconnection System Impact Study Report

For Queue Project Q22-2094

Bayport 138 kV

April 24, 2024

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q22-2094. The Interconnection Customer proposes a 400.2 MW natural gas combined cycle facility interconnecting at the Bayport 138 kV substation in the New Jersey area. The request was received on May 24, 2022. The study evaluated system intact and single contingency conditions using the regional planning case, including 501 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 2 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$9,700,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

Monitored Facility	Contingency	Loading (%)	Rating (MVA)	DFAX (%)
Bayport / Sparks 138 kV	Mercury Marine / Bayport 138 kV	194.0	193	100.0
Mercury Marine / Bayport 138 kV	Bayport / Sparks 138 kV	180.5	207	100.0

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Sparks / Bayport 138 kV	Rebuild line with higher-capacity conductor	\$6,213,000	100% (\$6,213,000)
Mercury Marine / Bayport 138 kV	Rebuild line with higher-capacity conductor	\$3,487,000	100% (\$3,487,000)
	Total allocated to project		\$9,700,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.